Medical Image Analysis 106 (2025) 103765

Contents lists available at ScienceDirect

Medical Image Analysis

journal homepage: www.elsevier.com/locate/media

Collaborative surgical instrument segmentation for monocular depth estimation in minimally invasive surgery

Xue Li [a,b], Wenxin Chen [b,c], Xingguang Duan [a,b], Xiaoyi Gu [a,b,*], Changsheng Li [a,b,*]

[a] the School of Mechatronical Engineering, Beijing Institute of Technology, 30 Xueyuan Road, Haidian District, Beijing, 100081, China
[b] the Key Laboratory of Biomimetic Robots and Systems, Beijing Institute of Technology, 30 Xueyuan Road, Haidian District, Beijing, 100081, China
[c] the School of Medical Technology, Beijing Institute of Technology, 30 Xueyuan Road, Haidian District, Beijing, 100081, China

ARTICLE INFO

Keywords:
Surgical vision
Depth estimation
Instrument segmentation
Multi-task framework

ABSTRACT

Depth estimation is essential for image-guided surgical procedures, particularly in minimally invasive environments where accurate 3D perception is critical. This paper proposes a two-stage self-supervised monocular depth estimation framework that incorporates instrument segmentation as a task-level prior to enhance spatial understanding. In the first stage, segmentation and depth estimation models are trained separately on the RIS, SCARED datasets to capture task-specific representations. In the second stage, segmentation masks predicted on the dVPN dataset are fused with RGB inputs to guide the refinement of depth prediction.

The framework employs a shared encoder and multiple decoders to enable efficient feature sharing across tasks. Comprehensive experiments on the RIS, SCARED, dVPN, and SERV-CT datasets validate the effectiveness and generalizability of the proposed approach. The results demonstrate that segmentation-aware depth estimation improves geometric reasoning in challenging surgical scenes, including those with occlusions, specularities regions.

1. Introduction

The field of robotics holds the promise of revolutionizing surgical procedures, ushering in an era of enhanced safety, consistency, and minimally invasive interventions. With continuous advancements, robot-assisted systems are steadily progressing towards autonomous surgical operations. Chadebecq et al. (2020) and Duan et al. (2023). A core component of such systems is the image acquisition module, which typically captures 2D images or videos of surgical instruments and anatomical structures (Huang et al., 2022; Li and Peng, 2022). These visual data streams are critical for intraoperative decision-making, navigation, and situational awareness. However, extracting clinically meaningful information from surgical videos remains highly challenging, especially in the presence of tissue deformation, instrument occlusion, and varying illumination conditions.

Among the key perception tasks, scene depth estimation and instrument segmentation are both fundamental to understanding intraoperative environments. Depth estimation provides essential 3D information for surgical navigation and planning, while segmentation identifies and localizes surgical tools and anatomical structures. In recent years, deep learning-based models have made notable progress in both areas, particularly with convolutional architectures such as FCN (Long et al., 2016), U-Net (Ronneberger et al., 2015), DeepLab (Chen et al., 2017), SegNet (Badrinarayanan et al., 2017), and PSPNet (Zhao et al., 2017) demonstrating strong performance in instrument segmentation.

While stereo cameras are available in some robotic surgical platforms such as the da Vinci system, monocular depth estimation remains highly relevant and practically important in the context of minimally invasive surgery. First, many clinical procedures — including gastrointestinal endoscopy, bronchoscopy, and cystoscopy — rely solely on monocular video streams, where stereo-based approaches are not applicable. Second, monocular depth estimation offers a cost-effective, lightweight alternative in scenarios where stereo imaging is impractical due to hardware, calibration, or deployment constraints. Third, even in stereo-capable systems, stereo correspondence can fail in low-texture regions, under occlusions, or with specular reflections, where monocular methods can serve as a complementary fallback or provide semantic priors.

This direction has recently gained momentum in surgical vision research. For example, EndoDepth (Reyes-Amezcua et al., 2024), Surgical Depth Anything (Lou et al., 2024), and WS-SfMLearner (Lou and Noble, 2024) have demonstrated the feasibility of reliable monocular depth prediction in endoscopic settings using data-driven methods. This study follows the line of work and further explores how monocular depth estimation can be enhanced by incorporating task-level priors

* Corresponding authors.
E-mail addresses: xiaoyi900217@gmail.com (X. Gu), lics@bit.edu.cn (C. Li).

https://doi.org/10.1016/j.media.2025.103765
Received 23 July 2024; Received in revised form 29 July 2025; Accepted 10 August 2025
Available online 18 August 2025
1361-8415/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

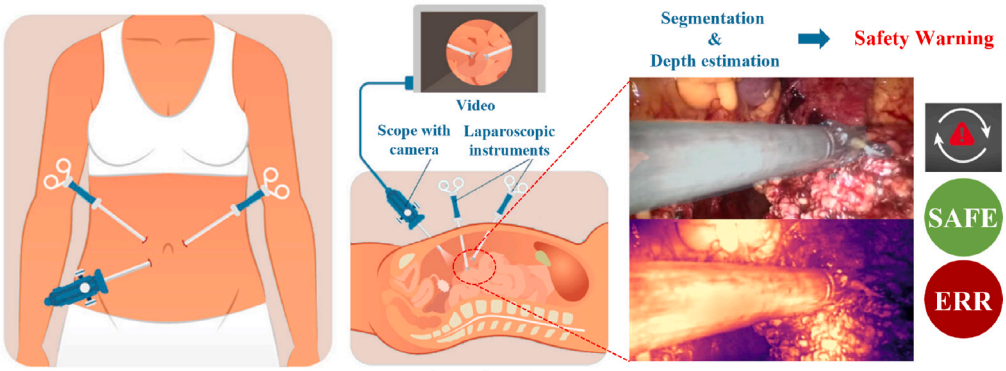

Fig. 1. Concept diagram of surgical instrument segmentation for monocular depth estimation in minimally invasive surgery.

— specifically, instrument segmentation — within a unified multi-task learning framework. This approach seeks to enable flexible and robust scene understanding across a wide range of surgical environments, beyond the limitations of stereo-only systems.

While these tasks have often been studied independently, multi-task learning has shown considerable promise in related domains. For example, in autonomous driving and outdoor scene understanding, integrating multiple perception tasks within a unified framework has been shown to improve efficiency and robustness. Such frameworks allow tasks to benefit from shared representations, and leveraging semantic priors can significantly enhance the accuracy of other tasks like depth estimation or flow prediction. Despite this progress in outdoor applications, multi-task learning strategies remain underexplored in surgical image analysis, partly due to the scarcity of annotated datasets and the complexity of surgical scenes.

Based on the aforementioned insights, a multi-task framework is constructed to jointly address surgical instrument segmentation and self-supervised monocular depth estimation. By leveraging shared representations and incorporating semantic priors, the proposed approach enhances depth prediction under the brightness constancy assumption. Experimental results demonstrate that integrating segmentation cues leads to more accurate and robust depth estimation, especially in the presence of challenging surgical conditions. Furthermore, the framework exhibits strong generalization capability even under limited data scenarios, indicating its potential applicability in real-world surgical systems (see Fig. 1).

The main contributions of this work are summarized as follows:

(1) Construction of a multi-task framework for instrument segmentation and monocular depth estimation. The model demonstrates competitive results in single-stage tasks individually. In two-stage tasks, the integration of both components aids in achieving a more comprehensive understanding of the scene.

(2) Further exploration delves into the impact of instrument segmentation results on the monocular depth estimation algorithm in two-stage tasks. Explore the impact of using instrument segmentation mask with different transparency on the depth estimation algorithm.

Across a spectrum of tasks, the proposed algorithm demonstrates superior performance over existing methods in terms of various evaluation metrics, including mean Intersection over Union (mIoU), Structural Similarity (SSIM), Peak Signal-to-Noise Ratio (PSNR), as well as traditional monocular depth estimation metrics such as Absolute Relative Error (Abs Rel), Root Mean Square Error (RMSE), and δ accuracy thresholds. These results affirm the validity and effectiveness of the model outlined.

2. Related works

2.1. Surgical instrument segmentation

Intraoperatively categorizing acquired images and classifying identified instances based on context and tools, can offer valuable surgical assistance, optimizing intraoperative decision-making. Traditional image segmentation tasks mainly include image features (Bhandari et al., 2019), machine learning (Rovera et al., 2023) and shape template matching algorithms (Kaur and Kaur, 2014). Nowadays, image segmentation algorithms based on deep learning are shining in various fields. Notably, FCN and U-Net have gained widespread recognition for their exceptional performance in medical image segmentation, spawning various derivative networks. In the EndoVis-2017 challenge, MIT introduced TernausNet (Kurmann et al., 2017), which features a decoder comprising multiple blocks connected to corresponding encoder blocks. Additionally, the University of Bern proposed an innovative approach leveraging cascaded FCN to enhance active stage results by reusing information from the preceding stage (Allan et al., 2019). Furthermore, recent research strives to enhance existing frameworks by integrating cutting-edge advances. This includes incorporating high-level modules into backbone networks or feature pyramid networks (Cerón et al., 2021), as well as introducing additional classification modules to extend current instance segmentation frameworks (Baby et al., 2023). The integration of attention mechanisms also plays a role in the development of instance-based segmentation, such as using dual attention modules to capture global information and using convolutional attention modules to adjust the global architecture (Cerón et al., 2021, 2022). In the EndoVis-2019 challenge, the CASIA_SRL team (Ross et al., 2021) actually adopted an attention module capable of capturing second-order statistics to resolve the semantic dependencies between pixels and capture global context, ultimately achieving instance segmentation.

2.2. Endoscopic image depth estimation

Classic depth estimation methods rely on multi-view geometry (Leonard et al., 2018), stereoscopic images (Marmol et al., 2018), and other traditional techniques. Yet, in surgical scenes, endoscopic images often exhibit sparse features and uneven keypoint distribution, which significantly affects the accuracy of these approaches.

In recent years, deep learning-based methods have gained attention. Due to the difficulty in acquiring real-world ground truth depth data, many studies rely on computer-generated (CG) datasets for training. This reliance, however, often limits the generalization of the models in clinical environments.

Recent approaches to depth estimation in both synthetic and real-world surgical images include unsupervised feedback networks (Hwang et al., 2021), Bayesian deep networks, and teacher–student frameworks (Rodriguez-Puigvert et al., 2022). Despite their progress, these methods still struggle to maintain spatial consistency across frames.

To address data limitations, Li et al. (2021) employed robot kinematics and single-frame inputs to construct a dense depth map (Robot-Depth), while Turan et al. (2018) developed an unsupervised odometry and depth estimation algorithm tailored for handheld and capsule endoscopes. Bardozzo et al. (2022) applied stereo-corrected disparity maps to improve monocular and stereo-based depth prediction.

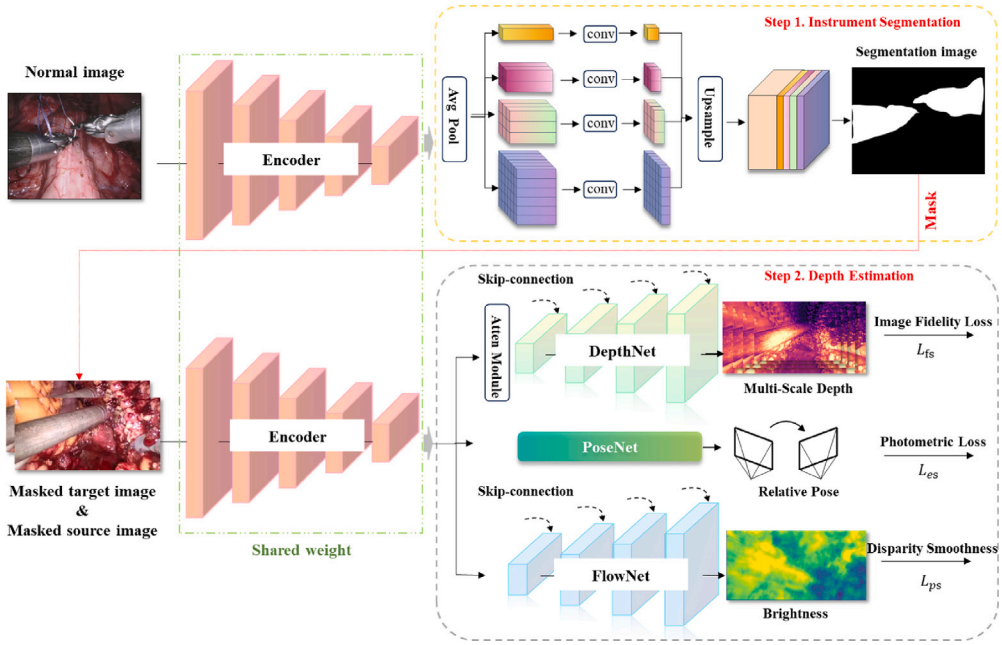

Fig. 2. Overview of a self-supervised monocular depth estimation framework for unified joint instrument segmentation.

In parallel, Recasens et al. (2021) introduced a monocular RGB-D reconstruction method by combining estimated camera poses with monocular image frames.

Nevertheless, many of these algorithms experience reduced performance in the presence of blurred object boundaries, low-texture regions, and illumination-induced brightness variations.

2.3. Multi-task learning

The utilization of shared encoders facilitates the learning of common features adaptable to diverse tasks, enhancing the consistency of feature learning and fostering a more comprehensive understanding of the input data. Extracting features only once, shared encoder can significantly reduce both inference time and overall system memory usage. Notably, Sense proposed a shared encoder that connects optical flow, stereo, occlusion, and semantic segmentation tasks, leveraging multi-task learning to achieve enhanced detection outcomes (Jiang et al., 2019). Similarly, MSDESIS proposed a shared encoder multi-task framework that combined disparity estimation and surgical segmentation for stereo endoscopic image analysis (Psychogyios et al., 2022). The medical image datasets utilized for distinct tasks exhibit considerable variation, lacking comprehensive public datasets that cater to the requirements of multi-task learning. As a consequence, there is limited research related to multi-task learning. The focus is to address how to enhance the adaptability of multi-task models within limited datasets and improve the performance of a single task.

3. Methods

This paper introduces a depth estimation network model that leverages instrument segmentation collaboration, which could accurately estimate intraoperative instruments and depth results. It is important to note that the proposed model can perform single-task or multi-task calculations based on specific requirements. A comprehensive exploration of the theoretical foundation and network structure will be provided as follows.

3.1. Preliminaries

Commonly employed self-supervised methods typically incorporate a DepthNet and a PoseNet. The DepthNet processes individual RGB images, producing estimated depth maps as its output. Simultaneously, the PoseNet is designed to operate on pairs of image frames, generating a pose transformation matrix with 6 degrees of freedom, encompassing both the rotation matrix (R) and translation matrix (T). Subsequently, upon obtaining depth estimates for each pixel in the target frame, a back-projection transformation is applied to project pixel points into the camera coordinate system. The pose estimation results are utilized to project the resulting three-dimensional coordinates onto the target image.

$$h\left(\mathbf{p}^{s\to t}\right) = [\mathbf{K}|\mathbf{0}]\mathbf{M}^{t\to s}\left[\mathbf{D}^{t}\mathbf{K}^{-1}h\left(\mathbf{p}^{t}\right)\right] \tag{1}$$

where $h\left(\mathbf{p}^{s\to t}\right)$ and $h\left(\mathbf{p}^{t}\right)$ represent the pixel coordinates of the source image and the target image, $\mathbf{K}$ is the intrinsic of the camera, $\mathbf{M}^{t\to s}$ is the output of PoseNet, represent the ego-motion from t to s, $\mathbf{D}^{t}$ is the depth of the target frame. The appearance difference between the target frame $I_t(p)$ and the synthesized frame $I_{s\to t}(p)$ supervises the entire training pipeline.

The above process relies on the assumption of consistent brightness between frames to impose constraints on the rigid flow. However, the specific characteristics of endoscopic scenes pose challenges, with constant illumination changes and non-Lambertian reflections severely disrupting L1 and SSIM. Therefore, this paper enhances the depth estimation network by adding auxiliary brightness estimation to its baselines. This enhancement assumes that the image intensity of I_t is affected by illumination changes and deformations associated with non-illuminated reflections. Furthermore, intraoperative procedures involve frequent movements of surgical instruments. Minimizing the impact of these instrument movements on depth estimation results is crucial for a more comprehensive understanding of the surgical scene. This contributes to improved depth perception and overall scene comprehension in dynamic surgical environments.

3.2. Network architecture

Fig. 2 illustrates the modular network framework, constructed around a shared encoder that extracts features at various scales from

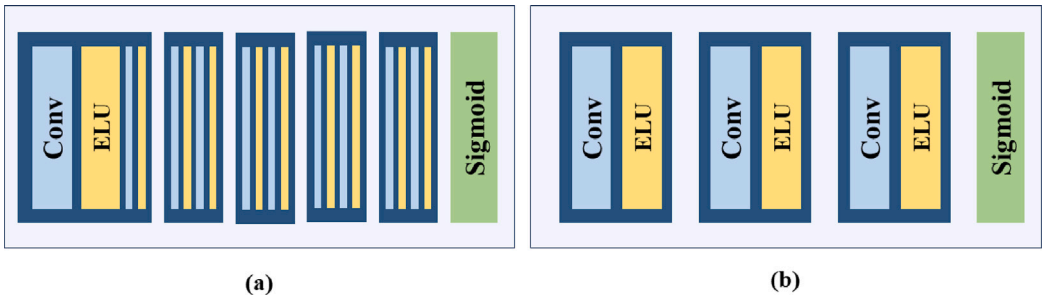

Fig. 3. Overview of the depth estimation algorithm decoder framework. (a) DepthNet and FlowNet decoder. (b) PoseNet decoder.

the input image. Corresponding decoders are designed for distinct tasks, using available multi-scale inputs for feature encoder prediction. In the segmentation task decoder, a pyramid pooling module is introduced, producing a binary segmentation mask with the same resolution as the input. For depth estimation tasks, the preliminaries proposed above are used. Fine-tuning training parameters allows both segmentation and depth estimation tasks to be tailored to entirely new domains.

3.2.1. Shared feature encoder

A ResNet18 backbone is employed as the encoder to extract hierarchical feature maps at five spatial resolutions, with channel dimensions of 16, 32, 64, 128, and 256, respectively. By incorporating features across different abstraction levels, the encoder enhances the network's capacity for robust feature learning and improves its generalization ability in multi-task settings.

3.2.2. Segmentation decoder

To address the potential information loss in various sub-regions, a pyramid pooling module is introduced for constructing a global scene prior. This module seamlessly integrates features from four distinct scales: 1×1, 2×2, 3×3, and 6×6. To maintain consistent global feature weights, a 1×1 convolution is applied to reduce the dimensionality back to the original 1/N. Subsequently, the low-dimensional feature map undergoes upsampling using bilinear interpolation to match the dimensions of the original feature map.

3.2.3. Depth decoder

To enhance the accuracy of the depth estimation model and mitigate the influence of illumination variations and non-Lambertian reflections on the final depth estimation results, a combined approach utilizing FlowNet, PoseNet, and DepthNet is employed. The brightness change information is obtained by the parallel prediction branch of FlowNet. The specific network composition is shown in Fig. 3.

Embedded within the network architecture is a dual attention mechanism, positioned at the output features of the last stage of the encoder and the input features of the first stage from the DepthNet decoder. This mechanism enhances network accuracy by capturing contextual information in both the channel and spatial dimensions.

The channel attention branch first compresses the input feature map through convolution, batch normalization, and ReLU activation. Multi-scale descriptors are then extracted using Spatial Pyramid Pooling (SPP) to capture contextual information across different receptive fields. These descriptors are processed by a lightweight Multi-Layer Perceptron (MLP) to compute channel-wise attention weights, which are applied to the original features after sigmoid activation to emphasize discriminative channels.

The spatial attention branch applies a set of multi-scale convolutional operations to the input, each capturing spatial dependencies at different scales. The outputs are passed through non-linear activation and combined through learnable weighting to generate the final spatial attention map, which highlights salient regions in the feature space.

The final output is produced by concatenating the attention-enhanced features and the original input via a residual connection, followed by convolution. This attention-guided fusion effectively improves the model's ability to handle texture-deprived areas, occlusions, and illumination variations commonly encountered in surgical video analysis.

FlowNet comprises five convolutional layers, with a convolutional operation sequence of $C_e256 - C_e128 - C_e64 - C_e32 - C_e16 - C_s3$. Skip connections are utilized to facilitate information exchange between the ith layer and the $n-i$-th layer, where n represents the total number of layers and $i \in 0, 1, 2, 3$. Moreover, the network incorporates a detailed weight initialization strategy to optimize training stability and improve overall performance.

3.3. Training details

The method consists of two-stage of training, the loss functions of the surgical instrument segmentation and the depth estimation will be introduced in detail.

3.3.1. Instrument segmentation

At this stage, the Robotics Instrument Segmentation (RIS) dataset obtained from the MICCAI 2017 EndoVis challenge is used. The task involved binary classification, distinguishing between foreground and background areas.

To enhance the network's ability to focus on task-relevant regions, SimAM (Yang et al., 2021) is introduced estimates attention weights based on an energy function derived from neuroscience principles, without introducing additional parameters or structural overhead. By incorporating SimAM, spatial and channel information are effectively integrated to generate 3D attention weights based on an energy function. This mechanism facilitates the emphasis of informative features and enhances global contextual understanding during feature extraction. Additionally, the Exponential Linear Unit (ELU) (Clevert et al., 2015) activation function is adopted to accelerate training convergence and improve gradient flow stability, especially in early training phases.

In the first stage of surgical instrument segmentation tasks, a common challenge lies in the small proportion of foreground regions and the severe imbalance between foreground and background. In such imbalanced scenarios, the traditional cross-entropy loss is easily dominated by the overwhelming number of background pixels, which leads the model to overfit background features while neglecting critical foreground targets, thereby degrading the overall segmentation accuracy.

To effectively mitigate this issue, this paper adopts a combination of two loss functions — Dice Loss and Focal Loss — to establish a complementary mechanism that balances regional structure overlap optimization and pixel-level discrimination enhancement. This strategy leverages the strengths of both losses in handling sparse foregrounds and varying levels of sample difficulty.

Furthermore, to enhance the representational capacity of deep features during training, an auxiliary supervision branch is incorporated. Through a deep supervision strategy, auxiliary losses are imposed on intermediate layers of the network to facilitate gradient flow and improve the learning of shallow features. As the network depth increases,

the auxiliary loss contributes significantly to training stability and convergence speed.

The final segmentation loss is defined as follows:

$$\mathcal{L}_{\text{seg}} = \alpha \cdot \left(\mathcal{L}_{\text{Dice}}^{\text{main}} + \mathcal{L}_{\text{Focal}}^{\text{main}}\right) + \lambda \cdot \left(\mathcal{L}_{\text{Dice}}^{\text{aux}} + \mathcal{L}_{\text{Focal}}^{\text{aux}}\right) \tag{2}$$

The terms $\mathcal{L}_{\text{Dice}}^{\text{main}}$ and $\mathcal{L}_{\text{Focal}}^{\text{main}}$ represent the Dice Loss and Focal Loss computed from the main output branch of the network, while $\mathcal{L}_{\text{Dice}}^{\text{aux}}$ and $\mathcal{L}_{\text{Focal}}^{\text{aux}}$ are calculated from the auxiliary supervision branch. The coefficients α and λ are weighting factors used to balance the contributions of the main and auxiliary losses. Following the setting used in Zhao et al. (2017), the auxiliary loss weight λ is set to 0.4 to enhance the training of intermediate layers without overwhelming the main supervision signal.

3.3.2. Depth estimation

At this stage, the focus is directed towards training the depth estimation model. Surgical instrument segmentation model is utilized to predict the devices within the dVPN dataset, which produced by Hamlyn Centre, Imperial College London (Ye et al., 2017) . The resulting output is a mask, overlaid on the original dataset to serve as input for the depth estimation model.

The existence of occluded regions in the scene can significantly increase the photometric loss, depending on both camera motion and scene structure. To address this issue, the photometric loss is computed by taking the minimum across a set of predefined photometric error metrics V, which helps to select the most reliable error at each pixel:

$$\mathcal{L}_p(p) = \min_{v \in V} pre_v\left(I_t(p), I_{s\to t}(p)\right) \tag{3}$$

where $pre_v(\cdot)$ denotes the photometric error under a specific metric v, typically computed as a weighted combination of SSIM and L1 losses. Minimizing over multiple metrics improves robustness to illumination changes and local inconsistencies.

To further suppress unreliable gradients caused by independently moving objects, a motion-consistency mask $\mu(p)$ is introduced. It filters out dynamic pixels by comparing localized errors with a global photometric baseline:

$$\mu(p) = \min_{v \in V} pre_v\left(I_t(p), I_{s\to t}(p)\right) < \min_{v \in V} pre_v\left(I_t, I_s\right) \tag{4}$$

This binary mask ensures that only consistent regions contribute to the optimization process.

Accordingly, the image fidelity loss is weighted by the motion mask:

$$\mathcal{L}_{fs} = \zeta \cdot \mathcal{L}_p(p) \tag{5}$$

where $\zeta = \mu(p)$ controls the spatial contribution of each pixel based on motion consistency.

To enhance the coherence of the predicted depth map and alleviate discontinuities often observed in textureless regions, an edge-aware smoothness term is integrated:

$$L_{es} = \left|\partial_x d_t^*\right| e^{-\left|\partial_x I_t\right|} + \left|\partial_y d_t^*\right| e^{-\left|\partial_y I_t\right|} \tag{6}$$

Here, d_t^* denotes the normalized inverse depth map, which is scaled using a normalization factor $\alpha = 0.85$ to avoid over-shrinking and maintain relative depth variations. The exponential edge term suppresses regularization in high-gradient (i.e., edge) regions, allowing depth discontinuities to align with image boundaries. To address the model's ability to perceive complex texture areas, relatively long distances, or low-resolution moving objects, a brightness loss function is introduced at this stage.

$$\begin{aligned}\mathcal{L}_{ps} = Mean[&\lambda * \text{SSIM}\left(I_t'(p), I_{s\to t}(p)\right) \\ &+ \mu * \text{MS}-\text{SSIM}\left(I_t'(p), I_{s\to t}(p)\right) \\ &+ \mu * \text{L}_1\left(I_t'(p), I_{s\to t}(p)\right)]\end{aligned} \tag{7}$$

The weighted combination of SSIM at different scales effectively enhances the sensitivity of the network to input images of varying resolutions and also contributes to improving low-level depth estimation. Finally, the complete loss function is formulated as follows:

$$L = L_{fs} + L_{ps} + \kappa L_{es} \tag{8}$$

3.4. Datasets

To address the specific requirements of surgical scene understanding and depth estimation, four publicly available datasets were utilized: the RIS dataset, the SCARED dataset, the SERV-CT dataset, and the dVPN dataset. Each dataset offers complementary characteristics tailored to different stages of training and evaluation.

3.4.1. Robotics instrument segmentation dataset

The Robotics Instrument Segmentation(RIS) Dataset (Allan et al., 2019) originates from the MICCAI 2017 EndoVis challenge and consists of high-resolution (1920 × 1080) laparoscopic videos from 10 porcine abdominal surgeries. The first 225 frames from 8 sequences were used for training, and the last 75 frames from the remaining 2 sequences were used for testing, totaling 3101 training and 300 testing images. All instrument annotations were converted to binary masks, disregarding part-level distinctions. To enhance model robustness under realistic surgical conditions, multiple augmentation strategies were applied using the `imgaug` library. Specifically, 600 image–mask pairs underwent motion blur with kernel sizes $\{30, 40, 50\}$ and angles $\{60°, 180°, 300°\}$. Additionally, 400 RGB images were augmented with artificial fog (intensity range: 0.1–0.4) to simulate surgical smoke, although their labels were not modified. Further, over 300 samples were processed using random rotations and horizontal flips. In total, the final training set included 3101 augmented RGB images with corresponding binary masks.

3.4.2. SCARED dataset

The SCARED dataset (Allan et al., 2021) comprises stereo images and ground-truth depth in the form of point clouds, captured from porcine cadaveric abdominal anatomy using the da Vinci Xi robotic system. It includes seven training sequences and two additional point cloud datasets. Due to synchronization and calibration issues in sequences 4 and 5, only sequences 1, 2, 3, 6, and 7 were used. Keyframes 1 and 2 from each selected sequence were used for training, while keyframe 3 was used for evaluation, yielding 10,065 training and 1000 evaluation samples, each with a resolution of 320 × 256.

3.4.3. SERV-CT dataset

The SERV-CT dataset (Eddie" Edwards et al., 2020) provides a high-fidelity benchmark for evaluating depth estimation algorithms in surgical scenes. It consists of 16 rectified stereo image pairs acquired from ex vivo porcine torso cadavers, along with ground-truth depth, disparity, and occlusion maps derived from cone-beam computed tomography (CBCT). Unlike synthetic or phantom-based datasets, SERV-CT captures both anatomical structures and the stereo endoscope itself, enabling precise spatial alignment and pixel-level supervision.

The dataset includes intrinsic and extrinsic calibration parameters, and covers a variety of tissue types and anatomical configurations. With an average depth error of approximately 2 mm and a disparity error of 2 pixels, SERV-CT serves as a reliable reference under challenging surgical conditions, including specular highlights, occlusions, and texture-sparse regions.

3.4.4. dVPN dataset

The dVPN dataset (Ye et al., 2017) consists of stereo video frames collected during real partial nephrectomy procedures using the da Vinci system. It includes 34,320 stereo image pairs for training and 7191 pairs for testing, sampled from 11 video sequences. All images are of resolution 192 × 384 and are synchronized with calibrated camera parameters. Both left and right images share identical intrinsic matrices provided by the dataset.

4. Experiments

4.1. Training strategy

To investigate the synergistic effect of instrument segmentation on monocular depth estimation, a two-stage training strategy was adopted, incorporating both tasks in a sequential manner.

4.1.1. Stage I: Independent task learning

In the first stage, the segmentation and depth estimation modules were trained separately on different datasets to ensure task-specific feature learning.

Instrument Segmentation. The segmentation model was trained on RIS dataset using a ResNet18 backbone and mixed-precision training. The Adam optimizer was employed with a cosine annealing learning rate scheduler. An auxiliary supervision branch was added to enhance feature representation. Specifically, the second and third encoder feature maps (with resolutions of 128×128 and 256×256) were extracted. The second feature map was processed by a dilated convolution layer with dilation rate $(2, 2)$ to expand the receptive field, followed by bilinear upsampling to match the spatial size of the third feature map. The two outputs were then fused via a Pyramid Pooling Module (PSP) for multi-scale context aggregation.

Depth Estimation. The depth estimation model was pre-trained on the SCARED dataset using a self-supervised photometric reconstruction framework. The Adam optimizer was used with $\beta_1 = 0.9$, $\beta_2 = 0.99$, an initial learning rate of 10^{-4}, and a batch size of 12 stereo pairs. The loss function weighted by hyperparameters $\lambda = 0.95$, and $\mu = 0.5$. This initialization facilitated stable convergence and improved accuracy on challenging surgical scenes.

4.1.2. Stage II: Mask-guided depth estimation

Collaborative Surgical Instrument Segmentation for Monocular Depth Estimation. In the second stage, the segmentation model trained in Stage I was used to generate instrument masks for the dVPN dataset. These predicted binary masks were superimposed onto the input RGB images as an additional channel, serving as prior knowledge to guide depth estimation. The modified input thus consisted of masked-RGB . The depth estimation model was fine-tuned on the dVPN dataset using the same optimizer and loss configuration as in Stage I. This stage aimed to leverage explicit instrument localization to improve depth prediction near occlusion boundaries and reflective regions.

4.2. Evaluation protocol

To comprehensively assess the performance of the proposed multi-task framework, distinct evaluation strategies were adopted for segmentation and depth estimation, each tailored to the characteristics of the respective datasets. The evaluation followed a two-stage procedure, corresponding to the sequential training pipeline.

4.2.1. Stage I: Independent task learning

Instrument Segmentation. Instrument segmentation was evaluated on the test split of the RIS dataset. The mean Intersection-over-Union (mIoU) was used as the primary metric, averaged across all 10 test sets.

Depth Estimation. The first stage of depth estimation evaluation was conducted using the SCARED dataset. Standard evaluation metrics were employed, including Absolute Relative Error (Abs Rel), Root Mean Square Error (RMSE), and δ-accuracy thresholds ($\delta < 1.25$, 1.25^2, and 1.25^3), enabling direct comparison with established monocular depth estimation methods.

4.2.2. Stage II: Mask-guided depth estimation

Collaborative Surgical Instrument Segmentation for Monocular Depth Estimation. The second evaluation stage utilized the dVPN dataset. Since this dataset does not provide ground truth depth, a view synthesis based evaluation method is used to evaluate the depth estimation performance.

The left-view RGB images were warped to the right-view perspective using the predicted depth, known camera intrinsics, and stereo baseline transformations. The synthesized right-view images were then compared to the ground-truth right images using SSIM and PSNR, which were computed only over valid projection regions. This approach minimizes penalties from occluded or invalid regions and emphasizes the model's capacity to generate geometrically consistent reconstructions.

5. Experiment and analysis

To comprehensively assess the effectiveness and generalizability of the proposed framework, the experiments were organized into two stages, each targeting different aspects of the model's performance. In Stage I, the instrument segmentation and monocular depth estimation modules were trained and evaluated independently. The segmentation task was conducted on the RIS dataset to validate the model's ability to localize surgical tools, while the depth estimation network was trained on the SCARED dataset using a self-supervised learning scheme. In this stage, the proposed method was compared against a range of state-of-the-art monocular depth estimation approaches, including SfM Learner (Zhou et al., 2017), SC-SfMLearner (Bian et al., 2019), and Endo-SfM (Ozyoruk et al., 2021), monodepth2 (Godard et al., 2017), AF-SfMLearner (Shao et al., 2022), and PC-Depth (Li et al., 2025), under standard depth evaluation metrics such as RMSE, RMSE log, and δ accuracy thresholds. In addition to in-domain evaluation, the model's cross-domain generalizability was examined by directly applying the pretrained depth estimation model to the SERV-CT dataset without any fine-tuning.

In Stage II, the trained segmentation network was used to generate instrument masks on the dVPN dataset, which were then integrated with RGB inputs to guide the refinement of depth prediction. Since dVPN lacks ground-truth depth annotations, the evaluation in this stage was conducted based on view synthesis quality, using SSIM and PSNR as the primary metrics to assess image reconstruction consistency.

A detailed presentation of the experimental results and analyses is provided in the following sections.

5.1. Stage I: Instrument segmentation experiment

Since the MICCAI 2017 Challenge data is used, the surgical segmentation algorithm is compared with the official statistical results (Allan et al., 2019). Table 1 shows the test results of the algorithms by the top ten teams on 10 test sets. Among these, three teams—UB, MIT, and TUM exceed an average mIoU of 0.85. The algorithm achieved an average mIoU of 0.869 across the 10 test sets. Notably, in test sets 5 and 9, the proposed algorithm outperformed the MIT and TUM teams, demonstrating a strong degree of competitiveness.

Fig. 4 illustrates the performance of the instrument segmentation algorithm on the test set. The image frames in the first three rows are consistent with those used in the official visualization detection images. The proposed algorithm effectively detects the Maryland Bipolar Forceps, the Prograsp Forceps, and the Large Needle Driver.

Notably, while BIT, U Washington, and U Alberta teams experienced significant misdetections in frame 247, the proposed method accurately identifies these instruments. The last row in Fig. 4 depicts two challenging scenarios: motion blur and smoke blur. The results demonstrate the proposed algorithm's strong robustness in handling these conditions.

In summary, the proposed algorithm exhibits high accuracy and effectively segments surgical instruments, even in difficult situations.

Table 1
Quantitative comparison of instrument segmentation results. MIT scored the highest in 4 datasets, the proposed algorithm scored the highest in 2 datasets, and UB scored the highest in 3 datasets.

	NCT	UB	BIT	MIT	SIAT	UCL	TUM	Delhi	UA	UW	OURS
Dataset1	0.784	0.807	0.275	**0.854**	0.625	0.631	0.760	0.408	0.413	0.337	0.819 (−0.035)
Dataset2	0.788	**0.806**	0.282	0.794	0.667	0.645	0.799	0.524	0.463	0.289	0.745 (−0.061)
Dataset3	0.926	0.914	0.455	**0.949**	0.897	0.895	0.916	0.743	0.703	0.483	0.898 (−0.051)
Dataset4	0.934	0.925	0.310	**0.949**	0.907	0.883	0.915	0.782	0.751	0.678	0.914 (−0.030)
Dataset5	0.701	0.740	0.220	0.862	0.604	0.719	0.810	0.528	0.375	0.219	**0.883**
Dataset6	0.876	0.890	0.338	**0.922**	0.843	0.852	0.873	0.292	0.667	0.619	0.878 (−0.044)
Dataset7	0.846	**0.930**	0.404	0.856	0.832	0.710	0.844	0.593	0.362	0.325	0.860 (−0.070)
Dataset8	0.881	0.904	0.366	**0.937**	0.513	0.517	0.895	0.562	0.797	0.506	0.919 (−0.018)
Dataset9	0.789	0.855	0.236	0.865	0.839	0.808	0.877	0.626	0.539	0.377	**0.883**
Dataset10	0.899	**0.917**	0.403	0.905	0.899	0.869	**0.909**	0.715	0.689	0.603	0.888 (−0.021)

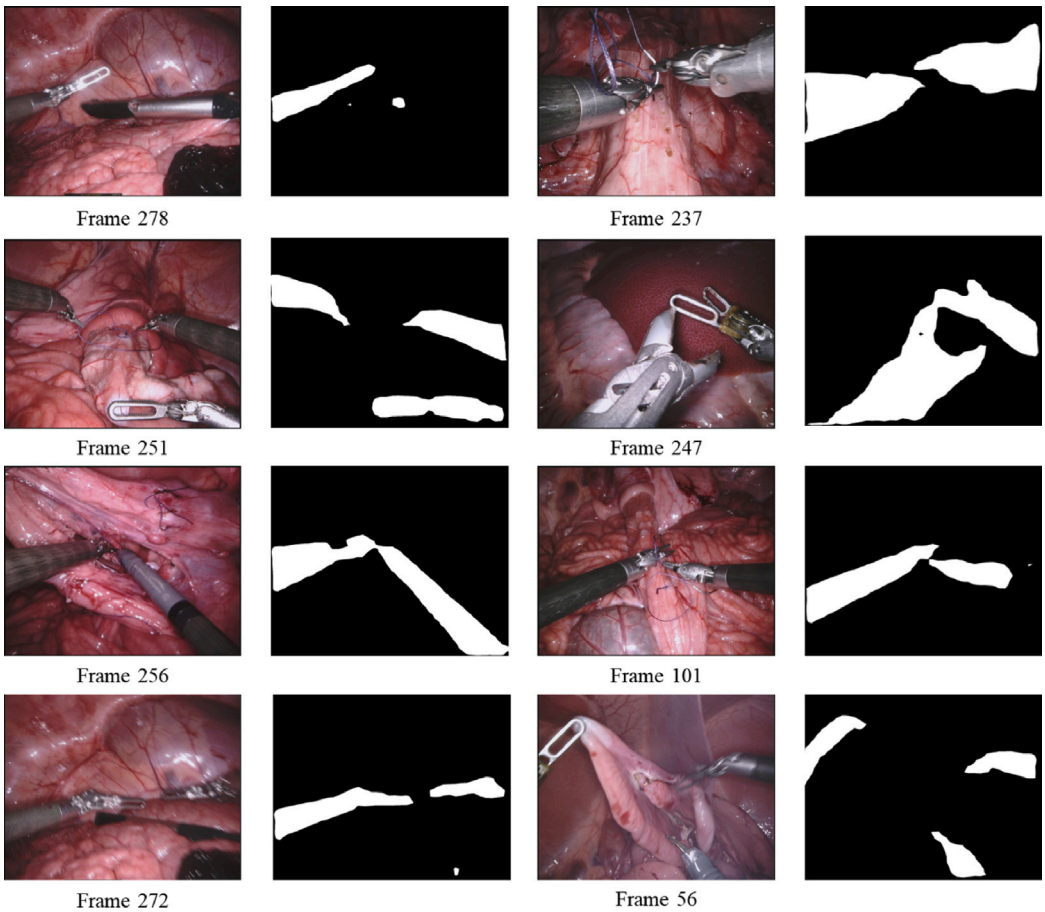

Fig. 4. The qualitative analysis of segmentation effects on selected images from the test set reveals distinct characteristics across Dataset 1, Dataset 3, Dataset 4, Dataset 7, Dataset 8, Dataset 9, and Dataset 10.

5.2. Stage I: Depth estimation experiment

5.2.1. Quantitative evaluation

The performance of various depth estimation algorithms was evaluated on the public standard SCARED dataset, and the experimental results are shown in Table 2. Among all the compared methods, the proposed method performed well in multiple key indicators, showing strong overall error control capabilities.

Compared with traditional self-supervised methods such as SfMLearner (RMSE = 6.969, but Abs Rel = 0.263, Sq Rel = 2.440) and SC-SfMLearner, this method improves error consistency while ensuring a low error margin. In addition, compared with the recently proposed Af-SfM and PC-Depth, this method also leads in most indicators. Af-SfM performs well in terms of accuracy ($\delta_3 = 0.994$), but its global alignment strategy is insufficient in restoring boundaries and details; although PC-Depth achieves a better result of 0.150 in RMSE log, its overall error (RMSE = 9.252) is still higher than this method, and its robustness in occlusion and motion areas is insufficient. In terms of accuracy, this method achieves the highest $\delta_1 = 0.868$, which is higher than PC-Depth (0.865) and Af-SfM (0.846), indicating that it has advantages in accurately predicting pixel-level depth.

To verify the effectiveness of each component, a step-by-step module ablation experiment was conducted on the SCARED dataset, and the results are listed in Table 3. In the baseline version without any guide module, the RMSE is 10.563 and the accuracy δ_1 is 0.804. After introducing the machine-guided attention mechanism, RMSE dropped to 10.168 and δ_1 increased to 0.822, indicating that semantic guidance can effectively enhance the model's perception of key areas. After further integrating the FlowNet refinement module, the performance continued to improve, and the final RMSE dropped to 8.771 and δ_1 reached 0.868, which is consistent with the complete model.

Table 2
Comparison of the proposed method to different methods on the SCARED dataset.

Methods	Errors				Accuracy		
	Abs Rel	Sq Rel	RMSE	RMSE log	$\delta 1$	$\delta 2$	$\delta 3$
SfM-Learner	0.263	2.440	6.969	0.299	0.542	0.866	0.967
SC-SfMLearner	0.292	3.044	7.702	0.331	0.503	0.823	0.950
Endo-SfM	0.290	3.065	7.893	0.330	0.492	0.818	0.948
Monodepth2	0.141	2.118	10.563	0.183	0.804	0.966	0.989
PC-Depth	**0.118**	1.497	9.252	**0.150**	0.865	**0.985**	**0.997**
Af-SfM	0.128	1.674	9.872	0.163	0.846	0.981	0.994
Ours	0.119	**1.481**	**8.771**	0.156	**0.868**	0.976	0.992

Table 3
The results of the ablation study on the SCARED dataset.

Methods	Errors				Accuracy		
	Abs Rel	Sq Rel	RMSE	RMSE log	$\delta 1$	$\delta 2$	$\delta 3$
baseline	0.141	2.118	10.563	0.183	0.804	0.966	0.989
baseline+Atten	0.136	2.000	10.168	0.179	0.822	0.965	0.988
baseline+FlowNet	**0.119**	**1.481**	**8.771**	**0.156**	**0.868**	**0.976**	**0.992**

To evaluate the cross-domain generalization ability of the model, the model trained on the SCARED dataset is directly tested on the SERV-CT dataset without any fine-tuning. As shown in Table 4, this method still maintains excellent performance on SERV-CT, achieving the lowest RMSE (11.017) and RMSE log (0.133), as well as the highest accuracy ($\delta_1 = 0.893$, $\delta_2 = 0.991$). In contrast, Af-SfM (RMSE = 12.513) and PC-Depth (RMSE = 15.963) show obvious performance degradation in cross-domain testing, and the accuracy also drops significantly.

5.2.2. Qualitative evaluation

Fig. 5 presents a comprehensive qualitative comparison of monocular depth estimation methods across two distinct datasets — SCARED (bottom row) and SERV-CT (top row) — which represent varying levels of visual complexity and clinical realism. For each dataset, the original image, predicted depth maps from representative methods, and zoomed-in regions of interest are displayed to emphasize differences in performance, particularly in challenging anatomical and surgical areas.

In the SCARED dataset, which features high-quality laparoscopic images with relatively clear texture and consistent lighting, traditional self-supervised approaches such as SfM-Learner and SC-SfMLearner struggle to preserve fine-grained anatomical structures. These methods often produce blurred or inconsistent depth predictions in low-texture areas. Although methods like Monodepth2 and PC-Depth effective in well-lit regions, fail to maintain reliable depth estimation in darker areas of the surgical scene. AF-SfM offers improved structure preservation, but this method achieves the most coherent and anatomically plausible depth reconstruction. As illustrated in the zoomed regions, the proposed model preserves subtle tissue folds with clear near-far gradients, demonstrating a strong capacity for structural awareness in standard laparoscopic views.

The SERV-CT dataset, representing real surgical scenes with challenging conditions such as occlusion, glare, and weak textures, is used to evaluate the generalization performance of the methods. Under this domain shift, prior methods experience a marked drop in estimation quality. SfM-Learner, SC-SfMLearner, and Endo-SfM exhibit noisy outputs, abrupt depth discontinuities, or overly flat estimations in critical regions.Monodepth2 and PC-Depth show partially improved consistency but remain vulnerable to illumination artifacts. AF-SfM demonstrates more stable depth geometry, yet struggles in some areas. Notably, the proposed approach consistently produces the most coherent and perceptually aligned depth maps across challenging areas, underscoring its enhanced generalization capability.

The visualization results in Fig. 6 show the predicted optical flow fields across temporally adjacent laparoscopic frames. In both cases, the optical flow estimation exhibits strong spatial consistency and anatomical alignment, validating the model's robustness under realistic surgical imaging conditions.

In the upper case, the predicted flow field is smooth and directionally coherent across the uniformly textured region in the peritoneal surface. This is particularly evident in the central abdominal area, where minimal depth variation exists and the illumination is relatively homogeneous. The absence of abrupt flow discontinuities in this region indicates that the model effectively regularizes motion estimation in low-gradient areas, a critical capability for handling texture-sparse anatomical surfaces.

In contrast, in the lower case, the scene contains complex folded intestinal structures with prominent vascular textures and strong specular highlights. Despite these challenges, the flow field maintains fine-grained detail, especially along organ boundaries and fold edges. These regions exhibit higher motion gradients, which reflect the model's structure-aware motion sensitivity, allowing it to distinguish between static background tissues and locally deforming organ surfaces. Furthermore, in darker regions (e.g., lower abdomen shadows), there is no noise amplification or collapse, suggesting a degree of photometric invariance.

The depth maps generated alongside the optical flow fields reveal consistent geometric structure across the surgical scene. In the upper example, the peritoneal surface appears relatively flat and continuous, with smooth depth gradients that correspond to the low-motion regions in the flow field. In the lower example, the folded anatomy of the intestine is captured with sharper depth transitions, which align well with the higher optical flow magnitudes near organ boundaries. These results suggest that the model maintains spatial coherence between depth and flow predictions. The alignment between motion discontinuities and depth gradients reinforces the geometric integrity of the model's representation.

5.3. Stage II: Collaborative surgical instrument segmentation for monocular depth estimation

5.3.1. Quantitative evaluation

Due to the absence of pixel-wise ground-truth depth annotations in the DPVN dataset, direct numerical evaluation of depth prediction accuracy is not feasible. To address this limitation, the SSIM is employed as a perceptual metric to indirectly assess the quality of monocular depth estimation. SSIM measures image similarity by evaluating structural, contrast, and luminance components, and has been widely used as a surrogate metric for evaluating reconstructed views in self-supervised depth learning frameworks.

In this study, depth predictions are evaluated by warping the source image to the target view using the predicted depth map, camera intrinsics, and known extrinsics. The resulting synthesized view is then compared to the actual target image using SSIM, calculated only in valid projected regions. This process reflects how well the predicted depth preserves structural integrity and enables accurate geometric transformations across views.

As reported in Table 5, the proposed method achieves the highest SSIM score of 92.05, outperforming both classical and recent monocular depth estimation baselines. Among the classical self-supervised methods, SfM-Learner (58.13), SC-SfMLearner (86.82), Endo-SfM (88.39),

Table 4
Generalization performance of different methods on the SERV-CT dataset.

Methods	Errors				Accuracy (Higher is better)		
	Abs Rel	Sq Rel	RMSE	RMSE log	$\delta 1$	$\delta 2$	$\delta 3$
SfM-Learner	0.138	3.486	15.574	0.183	0.820	0.952	0.993
SC-SfMLearner	0.118	2.052	12.922	0.152	0.864	0.986	0.999
Endo-SfM	0.107	1.790	11.972	0.141	0.884	0.985	0.999
Monodepth2	0.162	3.078	15.659	0.200	0.716	0.976	0.996
PC-Depth	0.153	3.346	15.963	0.197	0.746	0.958	0.993
Af-SfM	0.119	2.012	12.513	0.149	0.871	0.988	0.999
Ours	**0.101**	**1.561**	**11.017**	**0.133**	**0.893**	**0.991**	0.999

and Monodepth2 (71.47) show relatively limited performance in preserving structural details. In contrast, two recently proposed advanced methods, PC-Depth (56.99) and AF-SfM (85.70), offer improved results but still fall short of the proposed framework in both structural alignment and perceptual similarity.

This performance gain is largely attributed to the multi-task design of the framework, which integrates segmentation-guided priors, an attention-based module, and motion-aware refinement via optical flow. These components enable the model to focus on anatomically relevant features and maintain spatial coherence, even under challenging illumination and occlusion conditions.

Table 5
Quantitative comparison of depth estimate results on the dVPN Dataset.

Methods	SSIM	PSNR
SfM-Learner	58.13	21.10
SC-SfMLearner	86.82	27.53
Endo-SfM	88.39	28.17
Monodepth2	71.47	23.78
PC-Depth	56.99	21.76
Af-SfM	85.70	27.48
Ours	92.05	29.96

5.3.2. Qualitative evaluation

Fig. 7 presents qualitative comparisons of monocular depth estimation results on the DPVN dataset. The first row shows representative source frames from several laparoscopic procedures, characterized by challenging visual conditions including strong specular reflections, blood stains, and surgical instruments. Subsequent rows display the predicted depth maps generated by various baseline methods and the proposed model.

Among the classical methods, SfM-Learner and SC-SfMLearner produce overly smooth and flattened depth maps, failing to capture fine-grained anatomical structures. Their predictions lack depth contrast and suffer from structural collapse, particularly in texture-poor or overexposed regions. Endo-SfM slightly improves geometric detail but still exhibits inconsistent boundaries and low depth variation.

Monodepth2, though improved in visual sharpness, fails to maintain structural continuity across organ surfaces or capture fine-grained anatomical structures. PC-Depth, as a recent method, demonstrates stronger contrast but shows severe over-segmentation effects and depth fragmentation. AF-SfM also captures better depth variation but produces blurred contours and unstable depth layering in cluttered scenes.

In contrast, the proposed method yields more coherent and perceptually accurate depth maps. Anatomical boundaries are better preserved, with smooth near-to-far transitions and clearer structural gradients. Even in regions affected by surgical instruments, blood, or lighting variation, the predictions remain stable and consistent.

5.3.3. Ablation study on the effect of instrument segmentation masks

To investigate the effect of instrument segmentation on the accuracy of monocular depth estimation, a set of ablation experiments was conducted using the DPVN dataset. Specifically, segmentation masks of surgical instruments were overlaid onto the training images with varying levels of transparency: 0.0 (no mask), 0.1, 0.3, and 0.5. This masking procedure serves to introduce structural priors during training while allowing the network to focus on the geometric relationship between surgical tools and surrounding tissues.

Fig. 8 presents both quantitative results and qualitative visualizations. From the left panel, it can be observed that as the transparency level increases (i.e., the mask becomes more visually prominent), the instrument region becomes more distinguishable from the surrounding tissue. While masks with low transparency (e.g., 0.1) introduce minimal visual disturbance, higher opacity masks (e.g., 0.5) lead to visibly altered images, especially around the instrument boundaries.

6. Discussion

This study presents a self-supervised monocular depth estimation framework tailored for minimally invasive surgery, addressing key challenges such as specular highlights, occlusions, and texture sparsity in surgical scenes. To enhance robustness and structural awareness, the method introduces a two-stage training strategy that leverages segmentation-aware priors. In the first stage, instrument segmentation and depth estimation are trained independently to ensure stable task-specific representation learning. In the second stage, segmentation masks are fused with RGB images to guide the depth estimation model, allowing the network to focus on semantically relevant and geometrically informative regions.

Extensive experiments conducted on three datasets — SCARED, SERV-CT, and dVPN — demonstrate that the proposed framework consistently outperforms existing state-of-the-art methods across various evaluation metrics, including RMSE, SSIM, and δ accuracy thresholds. In particular, the method shows notable improvements in preserving fine-grained structures, producing smooth near-to-far transitions, and maintaining robustness in challenging regions such as reflective surfaces and occluded tissues. On the SERV-CT dataset, the model achieves competitive results without additional fine-tuning, confirming its strong cross-domain generalization capability.

The performance gains can be attributed to several core design features. The two-stage training strategy allows task-specific feature learning in the first stage and integrates semantic priors in the second stage, which guides the depth network to focus on geometrically informative areas. Unlike Af-SfM, which suffers from blurry boundaries under variable lighting, the proposed method achieves higher geometric fidelity through explicit spatial guidance. Compared with PC-Depth, which smoothes out important details in dark or specular areas, the proposed method is more robust in environments with instrumental interference. Classical photometric loss-based methods such as SfM-Learner and SC-SfMLearner are sensitive to brightness inconsistencies and struggle in low-texture areas, whereas the proposed model remains robust under such conditions.

A distinctive strength of the framework lies in its modular flexibility. The instrument segmentation and depth estimation modules can be deployed independently for specific clinical tasks, or jointly for enhanced performance. Segmentation can be applied to facilitate tool localization or robotic tracking, while depth estimation may support 3D reconstruction even in the absence of semantic inputs. When

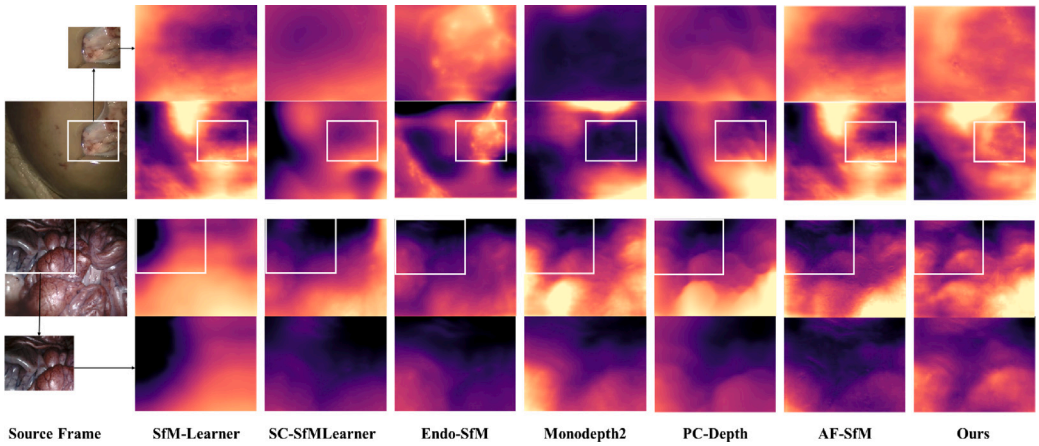

Fig. 5. Qualitative comparison of monocular depth estimation results on the SERV-CT (top) and SCARED (bottom) datasets. For each dataset, the source frame and predicted depth maps from various methods are shown, including classical baselines and recent approaches. White boxes denote regions of interest, which are further zoomed in to highlight local structural differences.

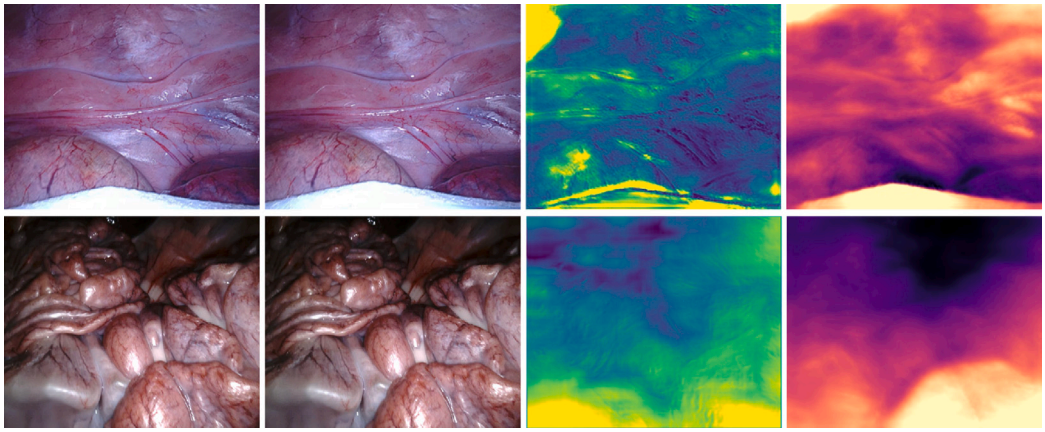

Fig. 6. Visualization of monocular depth estimation and flow prediction results on the SCARED Dataset. Each row displays a source frame, a temporally adjacent target frame, the predicted optical flow, and the corresponding monocular depth map.

combined, the segmentation-aware depth estimation offers enhanced geometric reasoning in semantically critical regions. This property opens up promising directions for future research, particularly in estimating relative depth between instruments and surrounding tissues. Such capability is of great value in narrow surgical spaces, where accurate safe-distance estimation is essential to avoid collisions and guide robotic interventions more safely and effectively.

Despite the promising results, the current approach has several limitations. The segmentation module is restricted to binary classification and does not differentiate among various instrument types, which limits the richness of the semantic priors. Additionally, while the model performs well at capturing large-scale geometric structures, it may struggle with very fine anatomical details, such as thin vessels or small polyps, especially in low-contrast conditions.

Future work will focus on addressing these limitations by extending the segmentation module to multi-class settings, incorporating temporal modeling to improve consistency across video frames, and developing self-supervised learning strategies that dynamically adjust loss weighting based on scene content. Such enhancements are expected to further improve the model's adaptability and generalization to complex, real-world surgical scenarios.

7. Conclusion

Given the challenges in obtaining accurate depth measurements in laparoscopic surgery, self-supervised depth estimation methods are gaining popularity. While previous methods have made considerable progress in addressing issues such as irregular lighting, blood stains, and tissue deformation, they still struggle with accuracy when surgical instruments are present in the field of view. Reflections from instruments and the significant area can significantly reduce the accuracy of existing algorithms.

To mitigate the impact of instrument on depth estimation accuracy, a novel approach that combines monocular depth estimation with instrument segmentation is introduced. By overlaying instrument segmentation results as masks onto the depth estimation dataset, the model gains a more comprehensive understanding of the scene and enhances its ability to perceive surgical instruments. Extensive experiments conducted on four datasets demonstrate the robust segmentation performance and accurate depth estimation achieved by the model.

In future research, there is a focus on expanding the capabilities of the model by incorporating additional tasks, such as assisting surgeons

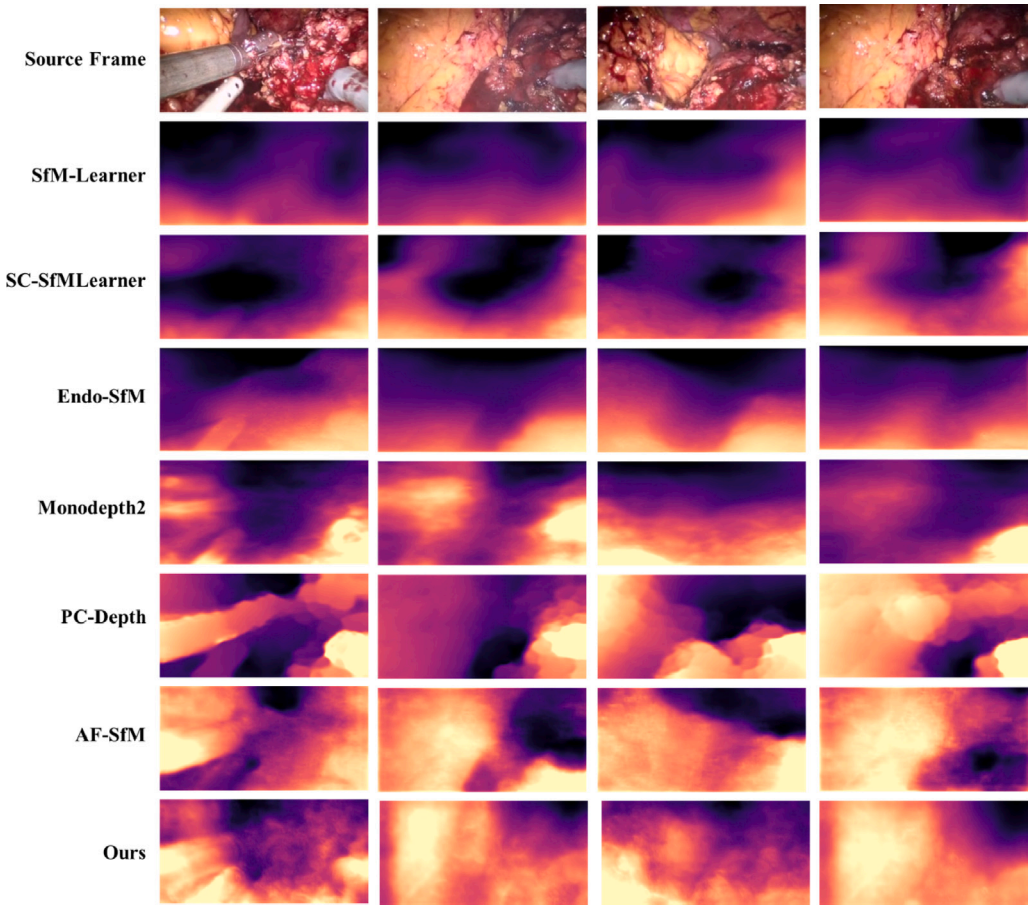

Fig. 7. Visualized results of experiments by superimposing the segmentation results of surgical instruments onto the original image to train a depth estimation model.

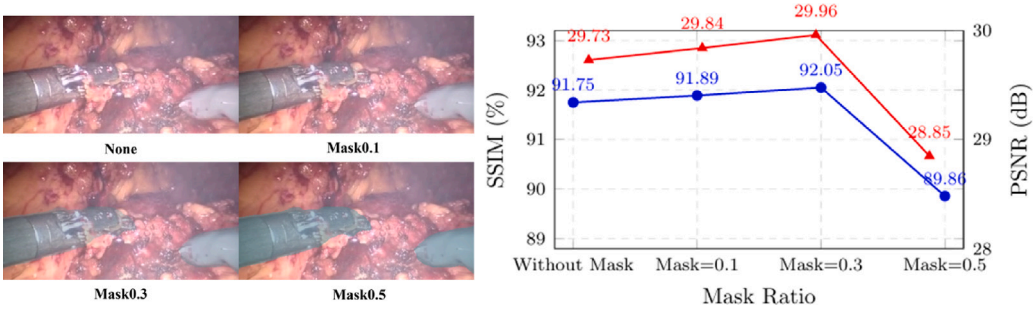

Fig. 8. Ablation study on the effect of instrument masks with varying levels of transparency. The left panel shows sample reconstructed images with different mask transparency levels applied to the instrument region. From top to bottom, the transparency increases. The right panel presents the corresponding quantitative results in terms of SSIM (%) and PSNR (dB), illustrating the impact of different mask ratios on reconstruction quality.

in determining safe distances between instruments and tissues during procedures. Furthermore, there are plans to enhance instrument segmentation by transitioning from simple two-category classification to a more nuanced multi-category classification, thereby further enhancing safety measures for autonomous surgery conducted by robotic systems.

CRediT authorship contribution statement

Xue Li: Writing – review & editing, Writing – original draft. **Wenxin Chen:** Data curation. **Xingguang Duan:** Project administration. **Xiaoyi Gu:** Supervision, Methodology. **Changsheng Li:** Methodology, Conceptualization.

Funding

This work was supported by National Key Research and Development Program, the Ministry of Science and Technology (MOST) of China (No. 2023YFB4706000), and the National Natural Science Foundation of China (No. 62373054)

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Allan, M., Mcleod, J., Wang, C., Rosenthal, J., Hu, Z., Gard, N., Eisert, P., Fu, K., Zeffiro, T., Xia, W., et al., 2021. Stereo correspondence and reconstruction of endoscopic data challenge. arxiv arXiv preprint arXiv:2101.01133.

Allan, M., Shvets, A., Kurmann, T., Zhang, Z., Duggal, R., Su, Y.-H., Rieke, N., Laina, I., Kalavakonda, N., Bodenstedt, S., et al., 2019. 2017 robotic instrument segmentation challenge. arXiv preprint arXiv:1902.06426.

Baby, B., Thapar, D., Chasmai, M., Banerjee, T., Dargan, K., Suri, A., Banerjee, S., Arora, C., 2023. From forks to forceps: A new framework for instance segmentation of surgical instruments. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision. pp. 6191–6201.

Badrinarayanan, V., Kendall, A., Cipolla, R., 2017. Segnet: A deep convolutional encoder–decoder architecture for image segmentation. IEEE Trans. Pattern Anal. Mach. Intell. 39 (12), 2481–2495.

Bardozzo, F., Collins, T., Forgione, A., Hostettler, A., Tagliaferri, R., 2022. Stasis-net: A stacked and siamese disparity estimation network for depth reconstruction in modern 3d laparoscopy. Med. Image Anal. 77, 102380.

Bhandari, A.K., Kumar, I.V., Srinivas, K., 2019. Cuttlefish algorithm-based multilevel 3-d otsu function for color image segmentation. IEEE Trans. Instrum. Meas. 69 (5), 1871–1880.

Bian, J., Li, Z., Wang, N., Zhan, H., Shen, C., Cheng, M.-M., Reid, I., 2019. Unsupervised scale-consistent depth and ego-motion learning from monocular video. Adv. Neural Inf. Process. Syst. 32.

Cerón, J.C.A., Chang, L., Ruiz, G.O., Ali, S., 2021. Assessing yolact++ for real time and robust instance segmentation of medical instruments in endoscopic procedures. In: 2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society. EMBC, IEEE, pp. 1824–1827.

Cerón, J.C.Á., Ruiz, G.O., Chang, L., Ali, S., 2022. Real-time instance segmentation of surgical instruments using attention and multi-scale feature fusion. Med. Image Anal. 81, 102569.

Chadebecq, F., Vasconcelos, F., Mazomenos, E., Stoyanov, D., 2020. Computer vision in the surgical operating room. Visc. Med. 36 (6), 456–462.

Chen, L.-C., Papandreou, G., Kokkinos, I., Murphy, K., Yuille, A.L., 2017. Deeplab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs. IEEE Trans. Pattern Anal. Mach. Intell. 40 (4), 834–848.

Clevert, D.-A., Unterthiner, T., Hochreiter, S., 2015. Fast and accurate deep network learning by exponential linear units (elus). arXiv preprint arXiv:1511.07289.

Duan, X., Xie, D., Zhang, R., Li, X., Sun, J., Qian, C., Song, X., Li, C., 2023. A novel robotic bronchoscope system for navigation and biopsy of pulmonary lesions. Cyborg Bionic Syst. 4.

Eddie'' Edwards, P., Psychogyios, D., Speidel, S., Maier-Hein, L., Stoyanov, D., 2020. Serv-ct: A disparity dataset from ct for validation of endoscopic 3d reconstruction. arXiv e-prints, pp. arXiv–2012.

Godard, C., Mac Aodha, O., Brostow, G.J., 2017. Unsupervised monocular depth estimation with left–right consistency. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 270–279.

Huang, B., Nguyen, A., Wang, S., Wang, Z., Mayer, E., Tuch, D., Vyas, K., Giannarou, S., Elson, D.S., 2022. Simultaneous depth estimation and surgical tool segmentation in laparoscopic images. IEEE Trans. Med. Robot. Bionics 4 (2), 335–338.

Hwang, S.-J., Park, S.-J., Kim, G.-M., Baek, J.-H., 2021. Unsupervised monocular depth estimation for colonoscope system using feedback network. Sensors 21 (8), 2691.

Jiang, H., Sun, D., Jampani, V., Lv, Z., Learned-Miller, E., Kautz, J., 2019. Sense: A shared encoder network for scene-flow estimation. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 3195–3204.

Kaur, D., Kaur, Y., 2014. Various image segmentation techniques: a review. Int. J. Comput. Sci. Mob. Comput. 3 (5), 809–814.

Kurmann, T., Marquez Neila, P., Du, X., Fua, P., Stoyanov, D., Wolf, S., Sznitman, R., 2017. Simultaneous recognition and pose estimation of instruments in minimally invasive surgery. In: Medical Image Computing and Computer-Assisted Intervention-MICCAI 2017: 20th International Conference, Quebec City, QC, Canada, September 11-13, 2017, Proceedings, Part II 20. Springer, pp. 505–513.

Leonard, S., Sinha, A., Reiter, A., Ishii, M., Gallia, G.L., Taylor, R.H., Hager, G.D., 2018. Evaluation and stability analysis of video-based navigation system for functional endoscopic sinus surgery on in vivo clinical data. IEEE Trans. Med. Imaging 37 (10), 2185–2195.

Li, S., Lin, W., Xiang, Q., Tu, Y., Asu, S., Li, Z., 2025. Unsupervised photometric-consistent depth estimation from endoscopic monocular video. In: Proceedings of the AAAI Conference on Artificial Intelligence. Vol. 39, pp. 4923–4931, no. 5.

Li, B., Lu, B., Lu, Y., Dou, Q., Liu, Y.-H., 2021. Data-driven holistic framework for automated laparoscope optimal view control with learning-based depth perception. In: 2021 IEEE International Conference on Robotics and Automation. ICRA, IEEE, pp. 12366–12372.

Li, R., Peng, B., 2022. Implementing monocular visual-tactile sensors for robust manipulation. Cyborg Bionic Syst..

Long, J., Shelhamer, E., Darrell, T., 2016. Fully convolutional networks for semantic segmentation. IEEE Trans. Pattern Anal. Mach. Intell. 39 (2), 640–651.

Lou, A., Li, Y., Zhang, Y., Noble, J., 2024. Surgical depth anything: Depth estimation for surgical scenes using foundation models. arXiv preprint arXiv:2410.07434.

Lou, A., Noble, J., 2024. Ws-sfmlearner: self-supervised monocular depth and ego-motion estimation on surgical videos with unknown camera parameters. In: Medical Imaging 2024: Image-Guided Procedures, Robotic Interventions, and Modeling. Vol. 12928, SPIE, pp. 119–127.

Marmol, A., Corke, P., Peynot, T., 2018. Arthroslam: Multi-sensor robust visual localization for minimally invasive orthopedic surgery. In: 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems. IROS, IEEE, pp. 3882–3889.

Ozyoruk, K.B., Gokceler, G.I., Bobrow, T.L., Coskun, G., Incetan, K., Almalioglu, Y., Mahmood, F., Curto, E., Perdigoto, L., Oliveira, M., et al., 2021. Endoslam dataset and an unsupervised monocular visual odometry and depth estimation approach for endoscopic videos. Med. Image Anal. 71, 102058.

Psychogyios, D., Mazomenos, E., Vasconcelos, F., Stoyanov, D., 2022. Msdesis: Multitask stereo disparity estimation and surgical instrument segmentation. IEEE Trans. Med. Imaging 41 (11), 3218–3230.

Recasens, D., Lamarca, J., Fácil, J.M., Montiel, J., Civera, J., 2021. Endo-depth-and-motion: Reconstruction and tracking in endoscopic videos using depth networks and photometric constraints. IEEE Robot. Autom. Lett. 6 (4), 7225–7232.

Reyes-Amezcua, I., Espinosa, R., Daul, C., Ochoa-Ruiz, G., Mendez-Vazquez, A., 2024. Endodepth: A benchmark for assessing robustness in endoscopic depth prediction. In: MICCAI Workshop on Data Engineering in Medical Imaging. Springer, pp. 84–94.

Rodriguez-Puigvert, J., Recasens, D., Civera, J., Martinez-Cantin, R., 2022. On the uncertain single-view depths in colonoscopies. In: International Conference on Medical Image Computing and Computer-Assisted Intervention. Springer, pp. 130–140.

Ronneberger, O., Fischer, P., Brox, T., 2015. U-net: Convolutional networks for biomedical image segmentation. In: Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5-9, 2015, Proceedings, Part III 18. Springer, pp. 234–241.

Ross, T., Reinke, A., Full, P.M., Wagner, M., Kenngott, H., Apitz, M., Hempe, H., Mindroc-Filimon, D., Scholz, P., Tran, T.N., et al., 2021. Comparative validation of multi-instance instrument segmentation in endoscopy: results of the robust-mis 2019 challenge. Med. Image Anal. 70, 101920.

Rovera, G., Grimaldi, S., Oderda, M., Finessi, M., Giannini, V., Passera, R., Gontero, P., Deandreis, D., 2023. Machine learning ct-based automatic nodal segmentation and pet semi-quantification of intraoperative 68ga-psma-11 pet/ct images in high-risk prostate cancer: A pilot study. Diagnostics 13 (18), 3013.

Shao, S., Pei, Z., Chen, W., Zhu, W., Wu, X., Sun, D., Zhang, B., 2022. Self-supervised monocular depth and ego-motion estimation in endoscopy: Appearance flow to the rescue. Med. Image Anal. 77, 102338.

Turan, M., Ornek, E.P., Ibrahimli, N., Giracoglu, C., Almalioglu, Y., Yanik, M.F., Sitti, M., 2018. Unsupervised odometry and depth learning for endoscopic capsule robots. In: 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems. IROS, IEEE, pp. 1801–1807.

Yang, L., Zhang, R.-Y., Li, L., Xie, X., 2021. Simam: A simple, parameter-free attention module for convolutional neural networks. In: International Conference on Machine Learning. PMLR, pp. 11863–11874.

Ye, M., Johns, E., Handa, A., Zhang, L., Pratt, P., Yang, G.-Z., 2017. Self-supervised siamese learning on stereo image pairs for depth estimation in robotic surgery. arXiv preprint arXiv:1705.08260.

Zhao, H., Shi, J., Qi, X., Wang, X., Jia, J., 2017. Pyramid scene parsing network. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 2881–2890.

Zhou, T., Brown, M., Snavely, N., Lowe, D.G., 2017. Unsupervised learning of depth and ego-motion from video. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 1851–1858.